Integration by parts: the product rule in reverse

The product rule gives

$$d(uv) = u\,dv + v\,du.$$

Integrate and rearrange:

$$\boxed{\int u\,dv = uv - \int v\,du.}$$

Choose u to differentiate and dv to integrate.

Choose u and dv, then follow four steps

Aim to make $\int v\,du$ simpler than the original integral.

1. Split the integrand: identify u and dv.
2. Differentiate u: find du.
3. Integrate dv: find v.
4. Substitute into $\int u\,dv = uv - \int v\,du$.

Useful choices for u: $\ln x$, inverse trig functions,

Example 1: $\int xe^x\,dx$

Choose

$$u = x, \quad dv = e^x\,dx \qquad \Longrightarrow \qquad du = dx, \quad v =$$

Then

$$\begin{aligned}\int xe^x\,dx &= xe^x - \int e^x\,dx \\ &= \boxed{xe^x - e^x + C}.\end{aligned}$$

Example 2: $\int \ln x \, dx \qquad (x > 0)$

Think of the integrand as $(\ln x) \cdot 1$:

$$u = \ln x, \quad dv = dx \qquad \Longrightarrow \qquad du = \frac{1}{x}\,dx, \quad v =$$

Thus

$$\begin{aligned} \int \ln x \, dx &= x \ln x - \int x \frac{1}{x}\, dx \\ &= \boxed{x \ln x - x + C}. \end{aligned}$$

Definite integrals: keep the boundary term

$$\boxed{\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du}$$

For $\int_0^1 xe^x\,dx$, use $u = x$, $dv = e^x\,dx$:

$$\int_0^1 xe^x\,dx = [xe^x]_0^1 - \int_0^1 e^x\,dx$$

Practice: choose u and dv first

1. $\int x \cos x \, dx$
2. $\int x \ln x \, dx \qquad (x > 0)$
3. $\int_0^1 x e^{2x} \, dx$

Check indefinite answers by differentiation.

Answers are on the next slide.

Practice answers

1. $u = x,\ dv = \cos x\, dx$:

$$\int x \cos x\, dx = x \sin x + \cos x + C.$$

2. $u = \ln x,\ dv = x\, dx$:

$$\int x \ln x\, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C.$$

3. $u = x,\ dv = e^{2x}\, dx$: